ASSIGNMENTS

Assignment guide

Research Assignment 1

Learning goals, real information systems and the engineer in the generative AI era

Research 1 task

Part	Required evidence
Learning goal	One skill you want to gain and why it matters
Real system	Users, information, action and one failure risk
AI era	One responsibility engineers still hold when AI helps
Reasoning	A connection among your goal, the system and the responsibility

Connect one personal goal to one real system and one current engineering responsibility.

Research 1 evidence and submission

Keep

Sources, one concrete system example, your own explanation and one remaining question.

AI if used

Show a useful discussion excerpt or link, verify important claims and explain your own conclusion.

Submit

Slides, a short report or another clear format. Email submit.progmis@gmail.com. File name to be announced.

Assigned Week 1, shared Week 2, worth 5%.

Research Assignment 2

Functions and tests that can reject a plausible wrong rule

Research 2 task

Part	Required evidence
Need + function	Name the coordination need, inputs, returned output and purpose
Wrong rule	Describe one plausible incorrect implementation
Test	Give input and expected output that expose that mistake
Reasoning	Explain why the test is stronger than one normal example

A useful test distinguishes the stated rule from a believable mistake.

Research 2 evidence and submission

Keep

The function contract, wrong rule, test input, expected result, sources and your explanation.

Share

Bring one test to the short Week 3 exchange. Normal full marks do not require a presentation.

Submit

Use a clear format. Email submit.progmis@gmail.com. Exact file name to be announced.

Assigned Week 2, shared Week 3 Session 7, worth 5%.

Research Assignment 3

Input validation and a clear response to invalid data

Research 3 task

Part	Required evidence
Need + contract	Name the decision supported, field, type and valid range or form
Invalid case	Give one malformed, missing or impossible value
Response	Reject, skip or request correction with a clear reason
Test	Show expected state and message after the invalid case

Validation protects a contract and produces evidence a person can act on.

Research 3 evidence and submission

Keep

The contract, invalid example, expected response, state check, sources and your reasoning.

AI if used

Challenge the suggested validation rule with a counterexample and verify it against a source.

Submit

Use a clear format. Email submit.progmis@gmail.com. Exact file name to be announced.

Released after the midterm, shared Week 5 Session 13, worth 5%.

Research Assignment 4

Persistence, database records and facts that must survive a restart

Research 4 task

Part	Required evidence
Need + fact	Name who needs one fact after the program closes
Record	Propose an identifier and two fields with clear units
Retrieval	State one question the stored record must answer
Risk	Explain one problem caused by missing or overwritten history

A persistent record needs an identity, useful fields and a retrieval question.

Research 4 evidence and submission

Keep

One persistent fact, record sketch, retrieval question, failure risk, sources and reasoning.

Check

Distinguish a saved historical fact from a value calculated only for display.

Submit

Use a clear format. Email submit.progmis@gmail.com. Exact file name to be announced.

Assigned Week 5, shared Week 6 Session 16, worth 5%.

Research Assignment 5

Transactions, failure handling and state that stays trustworthy

Research 5 task

Part	Required evidence
Need + operation	Name the reliable action and its two related writes
Failure	Place a failure between the writes
Expected state	State what must remain unchanged after rollback
Verification	Name the query or comparison that proves the outcome

A failure test should inspect state, not only the displayed message.

Research 5 evidence and submission

Keep

The two-step operation, failure point, before and after state, verification and sources.

Explain

Show why a friendly error message alone cannot prove that rollback worked.

Submit

Use a clear format. Email submit.progmis@gmail.com. Exact file name to be announced.

Assigned Week 6, shared Week 7 Session 19, worth 5%.

Research Assignment 6

Verification of output, including AI-generated output

Research 6 task

Part	Required evidence
Need + source	Name the decision and choose its original message, record or fact
Output	Choose one automated or AI-generated interpretation
Checks	Verify fields, units, omissions and unsupported claims
Decision	Accept, correct or hold the output with a reason

Verification compares output with evidence and preserves uncertainty.

Research 6 evidence and submission

Keep

Source, output, check results, final decision, limitation and your own reasoning.

No paid AI

Use supplied output when needed. Full marks do not depend on a live model or subscription.

Submit

Use a clear format. Email submit.progmis@gmail.com. Exact file name to be announced.

Assigned Week 7, shared at the start of Week 8, worth 5%.

Individual midterm

Basic Python, common tests, a reproducible ZIP and individual explanation

Midterm programming task

Need + calculate

A coordinator needs one summary. Loop through every line and total containers and declared planned portions.

Apply the rule

Return hold before quantity rules. Reviewed totals below 20 are standard; at least 20 is team.

Return + explain

Return the three supplied fields, test them and explain what the provisional label does not prove.

Start with the coordination question, then use Weeks 1 to 4 Python to answer it.

Four common minimum tests

Case	Input	Expected
A Review gate	4 × 5, not reviewed	4 containers, 20 portions, hold
B Below	3 × 3, reviewed	3 containers, 9 portions, standard
C Equality	4 × 5, reviewed	4 containers, 20 portions, team
D Multi-line	2 × 3 and 4 × 4, reviewed	6 containers, 22 portions, team

Every submission must assert all three returned fields for every case.

Midterm ZIP, README and AI disclosure

ZIP

Include README.md, midterm.py and test_midterm.py. Extract and test the ZIP in a fresh folder.

README

State purpose, Python version, exact run commands, expected output, limitation and AI use.

AI disclosure

State no use or the stages used: planning, approach, drafting, review, testing, debugging or wording.

A third party should run the work by following README.md.

Proposed 30-point rubric

Criterion	Points	Evidence
Totals and units	6	Correct loop and totals
Review-first rule	6	Correct labels and equality
Common tests	6	Four reproducible cases
Structure	4	Clear function and names
README and ZIP	4	Third party can run it
Explanation and disclosure	4	Reasoning and AI statement

Full standard marks do not require AI or optional refactoring.

Midterm timing and individual explanation

1

Implement

Recommended 90-minute work limit after Session 11.

2

Submit ZIP

Use the course submission email. Deadline to be announced.

3

Demonstrate

Run a case, explain the loop and answer one question.

4

Optional bonus

Improve a visible design issue only if the cap is later announced.

Submit to submit.progmis@gmail.com. One examiner can assess at most 15 five-minute explanations in one 90-minute block.

Final project

A small verified improvement to the supplied Python and SQLite system

Final project sequence

1

Week 8

Pitch a need, select a small change and freeze two acceptance cases.

2

Week 9

Implement, exchange a peer test, revise and retest.

3

Week 10

Rehearse from clean data, demonstrate and hand over the work.

4

Individual check

Trace a result or make one approved small change.

Research closes at the start of Week 8. Weeks 9 and 10 focus on the project.

Final project requirements

Need and scope

Name the stakeholder, approved change, what stays outside scope and at least two acceptance cases.

Evidence

Show working code, a boundary or failure case, one peer finding, revision and retest.

Handoff

Provide clean starting data, run instructions, expected observations, recovery advice and one limitation.

A small reliable Python or SQL improvement can earn full standard credit.

Final project assessment evidence

Area	Evidence
Scope	Concrete need, approved change and acceptance cases
Implementation	Working change that preserves course rules
Verification	Acceptance, regression and boundary or failure evidence
Peer revision	Finding, change and retest
Handoff	Another person can run and understand the work
Individual understanding	Each member explains a judgment and test

The final project is proposed at 30%; the point split still requires confirmation.

Final project submission status

Keep

Code, fictional inputs, tests, peer evidence, README or run guide and individual explanation.

Demonstrate

Show baseline and improvement. Let a peer try an unseen input and explain one limitation.

Still pending

File name, dates, team allocation, presentation slots and rubric point split.

Submit to submit.progmis@gmail.com. Keep the complete package and a backup.

Further reading in English

1. ## Python Tutorial: defining functions

   https://docs.python.org/3/tutorial/controlflow.html#defining-functions

2. ## Python unittest documentation

   https://docs.python.org/3/library/unittest.html

Read one relevant section, then bring an example and a question.